

WITH CHANDRAYAAN-3, ISRO CHOSE A FAILURE-BASED APPROACH AND CLEARANCE FROM ALL TEAM HEADS BEFORE LAUNCH

MOX or Mission Operations Complex-2 in the heart of Peenya, a teeming hub of small-scale industries in Bengaluru, has an unimpressive PWD-type façade. The road leading up to it does have a fresh coat of tar. It was in preparation for Prime Minister Narendra Modi's visit on August 26 to congratulate Indian Space Research Organisation (ISRO) scientists for the superlative success of the Chandrayaan-3 mission to the moon. But it had to be put off till after his visit due to security reasons.

On the MOX porch are displayed real-size models of Vikram, India's now-famous moon lander, and its companion, the Pragyan rover. They were transported from the UR Rao Satellite Centre (URSC) at the other end of the city for the PM's benefit. Sheathed in a shiny gold-coloured thermal blanket, Vikram, with its four landing legs, is surprisingly tall—around 10 feet—but has a squat box-like appearance. Weighing 1,742 kilograms, landing it on the moon is like dropping a mini-SUV on the satellite's surface. Pragyan, the rover whose glistening aluminium wheels bear the imprint of India's national emblem, resembles an oversized skateboard. ISRO scientists and their families were busy taking selfies with the spacecraft replica before it was dismantled and returned to the URSC.

Inside MOX-2, the three mission control halls were a hive of activity when INDIA TODAY visited their premises on August 29. The Chandrayaan project team had to move to another set of computer consoles to make way for the Aditya L1 mission team that was preparing for the spacecraft's launch on September 2 to study the sun. That's just how busy ISRO is these days. On the high walls of the complex were giant computer screens that displayed dashboards of Vikram's health and Pragyan's movements. The rover literally crawls over the grey-black scarred lunar surface at a 'top speed' of one centimetre per second, with two front-loaded cameras acting as its eyes. The terrain is hazardous, full of deep craters; on that day, the rover had to reverse from one just eight metres away from the lander. But not before swivelling around and taking the very first photograph of Vikram on the lunar surface, an image destined to become a permanent symbol of India's prowess.

Back on Earth, hunched on one of the consoles is the unassuming 46-year-old Palanivel Veeramuthuvel, in a dark cotton long-sleeved shirt and fawn-coloured check pants, his salt-and-pepper hair combed neatly back. A native of Tamil Nadu, Veeramuthuvel studied in the Villupuram railway school before completing an engineer-